

1. Administrative & Study Identification

Full Study Title: Prospective Registry Investigating Maternal, Infant, and Lactation Outcomes in Anifrolumab Users **Official Title:** PRIMULA Lac (Prospective Registry Investigating Maternal, Infant, and Lactation Outcomes in Anifrolumab Users): The AstraZeneca Lactation Study for Anifrolumab **Short Title/Acronym:** PRIMULA Lac **Protocol Number:** D3461R00052 **NCT Number:** NCT06594068 **Posting ID:** 713214307004669893 **Sponsor/Company:** AstraZeneca **Investigational Product:** Anifrolumab (Saphnelo) **Phase of Development:** Phase IV **Trial Status:** RECRUITING **Disease Area:** Systemic Lupus Erythematosus (SLE) **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the presence and concentration of anifrolumab in human breast milk and serum (maternal and infant) among lactating individuals receiving anifrolumab therapeutically. **Secondary Objectives:** To evaluate the safety, potential exposure, and health outcomes of breastfed infants of mothers treated with anifrolumab.

2.2 Background & Rationale To address the gap in safety data regarding the transfer of anifrolumab to breast milk and its effects on breastfed infants. This study is being conducted as a Post Marketing Requirement (PMR) to fulfill FDA mandates, ensuring the safety and efficacy of anifrolumab for breastfeeding mothers with moderate/severe Systemic Lupus Erythematosus (SLE) and their infants.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Prospective Registry) **Control Type:** Single-arm **Blinding:** Open-label (No masking) **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 16 lactating individuals **Number of Sites:** Multiple (US/Global) **Estimated Study Start:** 29-Aug-2025 **Participant Duration:** Total duration of participation for each participant will be approximately 1 month.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

Maternal: 1. 18 years or older. 2. Signed informed consent to participate. 3. Diagnosis of moderate/severe SLE. 4. Ongoing treatment with anifrolumab. 5. Has reached or will reach steady state (~85 days postpartum, at least 3 consecutive previous doses during the post-partum period) with anifrolumab by the time of study Day 1 (pre-dose milk collection). 6. Established lactation in the index post-partum period (breastfeeding or pumping for at least 4 weeks at time of Day 1 visit to ensure mature milk production). 7. Willing to breastfeed or pump regularly during the study period to maintain milk supply and exclusively pump breast milk for the 24-hour period of breast milk collection on Day 1 post IV dose. 8. Plans to continue feeding infant breast milk at least throughout the duration of the study and is not weaning. 9. Must be exclusively breast milk-feeding their infant (or if not exclusively breast milk-feeding, not providing more than 1 supplemental bottle of formula per day) at the time of enrollment and throughout the study period. 10. Agrees to use only lanolin nipple cream during the sampling period.

Infant: 1. Gestational age at delivery ≥ 32 weeks. 2. Birthweight > 10th percentile. 3. Weight > 10th percentile at the time of enrollment.

4.2 Exclusion Criteria

Maternal: 1. Received any investigational compound or approved biologic or biosimilar within 30 days or 5 half-lives (whichever is longer) prior to enrollment in the study. 2. Diagnosis of lupus nephritis in the last 12 months. 3. History of breast implants, breast augmentation, or breast reduction surgery that significantly impacts breastfeeding or collection of milk from 1 or both breasts. 4. History of malignancy in the last 10 years. 5. History of mastectomy. 6. Evidence of mastitis or any other significant active infection at Day 1 (pre-dose).

Infant: 1. Any abnormality noted or clinically significant medical condition, including cardiac, pulmonary, and liver disease, glucose instability, or active infection at the time of screening that, in the opinion of the investigator, may make implementation of the protocol or interpretation of the trial difficult or would put the infant participant at risk by participating in the study.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Standard therapeutic dosing of anifrolumab (typically 300 mg IV every 4 weeks or 120 mg SC once weekly) as prescribed by the treating physician. **Route of Administration:** Intravenous (IV) or Subcutaneous (SC). **Duration of Treatment:** Aligned with standard-of-care clinical treatment for SLE.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant medications for SLE (e.g., antimalarials, oral corticosteroids) as prescribed. **Prohibited:** Use of other investigational monoclonal antibodies, biologics, or therapies conflicting with safety assessments within 30 days or 5 half-lives prior to enrollment.

6. Study Timeline & Visit Schedule

Visit ID	Timing	Key Activities/Procedures
Screening	Prior to Day 1	Informed Consent, Maternal/Infant Medical History, Eligibility Confirmation
Day 1	Day 1	Maternal serum collection (pre-dose & 0-4 hours post-dose); exclusive breast milk pumping for 24-hour period post IV dose
Interim	Days 2 to 29	Milk collection at a series of 14 timepoints total; Maternal serum collection on Day 12 and approx Day 29 (immediately preceding subsequent dose)
Day 30 (End of Study)	Approx Day 30	Infant serum collected following the next dose and after 24 hours of breastfeeding; Maternal and infant AE monitoring completion

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- 1. Area under the milk concentration-time curve during a dosing interval:** Derived using non-compartmental analysis methods from concentration-time data for all evaluable participants.
- 2. Area under the milk concentration-time curve from time 0 to last quantifiable concentration:** Assessed via non-compartmental analysis methods for PK characterization in lactating individuals.
- 3. Average milk concentration at steady state:** Calculated by dividing the "Area under the milk concentration-time curve during a dosing interval" by the dosing interval for anifrolumab.

7.2 Secondary Endpoints

- 1. Total amount of drug excreted in milk over 24 hours.**
- 2. Fraction of dose excreted in milk.**
- 3. Maternal serum pharmacokinetic (PK) concentrations.** (*Note: Maternal and infant adverse events (AEs) will also be actively collected for the duration of the study.*)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via patient interviews, diaries, and scheduled HCP visits for both mother and infant. Actively collected for the full duration (~1 month).

Serious Adverse Events (SAEs): Required reporting timeline, typically within 24 hours of site awareness.

Data Monitoring Committee (DMC): Internal safety tracking governed by AstraZeneca to fulfill FDA PMR requirements.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set and PK Population (evaluable participants providing concentration-time data). **Sample Size Justification:** Target $N = 16$ is powered adequately to meet FDA guidelines for post-marketing lactation pharmacokinetic estimations. **Handling of Missing Data:** Standard pharmacokinetic imputation methods for samples below the limit of quantification (BLQ). Non-compartmental analysis methods will be used for milk PK derivations.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard onsite and remote monitoring to ensure the integrity of the breast milk and serum sample collection chain. **Audit Trail:** Strict documentation of sample transport/storage, eCRF locking, and data clarification forms.

11. Study Identifiers & Governance

Primary ID: D3461R00052 **Registry Number:** NCT06594068 **Posting ID:** 713214307004669893 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** PRIMULA Lac **Official Regulatory Title:** PRIMULA Lac (Prospective Registry Investigating Maternal, Infant, and Lactation Outcomes in Anifrolumab Users): The AstraZeneca Lactation Study for Anifrolumab

12. Clinical Framework (Systemic Lupus Erythematosus Specific)

Primary Condition: Systemic Lupus Erythematosus (SLE) **Diagnosis Threshold:** Clinically confirmed moderate/severe SLE requiring therapeutic intervention **Medical Methodology:** Type 1 interferon receptor antagonist therapy (Anifrolumab) **Optimization Target:** Assessment of maternal-infant drug transfer and lactation safety profile

13. Study Procedures & Methodology

Management Pathway: Standard therapeutic pathway for SLE combined with registry sample collections **Education Protocol (HCP):** Training on proper breast milk and infant serum extraction, handling, and storage **Education Protocol (Patient):** Self-collection instructions for breast milk at home, including restrictions to use only lanolin nipple cream. **Quality Audit**

Mechanism: Laboratory sample viability tracking & maternal adherence to collection schedules

14. Observation & Follow-Up Schedule

Total Duration: Approximately 1 month **Onsite Visit Cadence:** Periodically aligned with scheduled SLE treatment infusions/ injections (Day 1, Day 12, Day 29, Day 30) **Remote Monitoring Frequency:** Regular interval calls for infant safety tracking **Checklist Review Interval:** Milk collection diary review at each scheduled visit

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				